Alexander Terentyev

Software Engineer

Antwerp, Belgium | +32485129110 | alex.terentyev.dev@gmail.com
Portfolio website: behavez.github.io | LinkedIn: linkedin.com/in/alexanderterentyev

Professional Summary

Software engineer with production experience shipping C++ and C# systems across Steam, Meta, and Pico platforms. Built and iterated on player engagement mechanics, AI behaviour trees, extensible systems, and rapid-prototype gameplay loops in live production codebases. Passionate about designing robust, maintainable systems that give designers creative control.

Awards & Recognition

GDWC Winter Awards | 2024 | Finalist, Best Hobby Game
(*Best Hobby Game in the GDWC Winter Awards 2024 (featured at 16:30 in the video)*
The Rookies | 2024 | Selected Entry
(*the rookies for the rookie awards 2024)*

Professional Experience

Gameplay Systems Engineer
AlterEyes – Hasselt, Belgium | Sep 2025 – Present

- ✓ Shipped player-facing gameplay systems for Color a Cube across Steam, Meta, and Pico platforms, enabling a multi-platform release with a single unified codebase by architecting extensible C++ and C# feature layers that required no platform-specific rewrites.
- ✓ Delivered a cross-platform achievement system supporting player engagement and retention mechanics across 3 storefronts, built as a reusable, maintainable C# library configurable by designers without engineering support.
- ✓ Improved developer iteration speed by building a modular cheat-menu debugging package, enabling engineers and designers to prototype and validate gameplay mechanics faster through rapid prototyping workflows.
- ✓ Tracked, triaged, and resolved production crashes using Sentry to maintain player experience quality post-release, reducing open critical issues by systematically analysing stack traces and shipping targeted fixes.
- ✓ Ported an existing production title to the Pico platform end-to-end, integration, testing, and release preparation, demonstrating technical excellence across multiple platforms under strict deadlines.
- ✓ Implemented AI gameplay behaviour systems for NPC characters using OOP design patterns to produce robust, maintainable decision-logic iterable by designers via modular component structures.
- ✓ Optimised StrikeMania (retro arcade time-attack game on Steam): pooling, allocation trimming and change-skipping renderers measured by an editor benchmark, plus graphics presets and display settings.

Software Developer – Placement Year
AlterEyes – Hasselt, Belgium | Sep 2025 – Jan 2026

- ✓ Accelerated new concept evaluation by rapidly prototyping gameplay mechanics in a production codebase, enabling the studio to make go/no-go architectural decisions before committing to full implementation.
- ✓ Ported an existing production project to Steam Deck, broadening the player base by resolving platform-specific input, rendering, and performance constraints through targeted optimisation passes.
- ✓ Built gameplay tools and systems with a designer-first API, reducing engineering handoffs by enabling game designers to configure and integrate systems independently.

- ✓ Maintained code quality and team velocity through structured branching, pull request reviews, and progress tracking via YouTrack — keeping iterative delivery on schedule against strict deadlines.

Featured Projects

Tron BattleTanks
C++ / SDL2 | github.com/BeHaVeZ/FMLEngine-Tron_BattleTanks

- ✓ Built a top-down tank game on an engine-style SDL2 architecture; implemented core gameplay systems including AI movement, collisions, audio, and full scene flow.
- ✓ Designed ECS-style game objects and reusable components (transform / render / collision) to keep gameplay code modular and maintainable, demonstrating extensible architectural decision-making.
- ✓ Implemented enemy AI movement using raycasts for wall detection with validated corner probes, producing smooth, responsive NPC behaviour without physics overhead.
- ✓ Built scene/state management (menu, gameplay, highscores) with clean lifecycle transitions and highscore persistence via file I/O with a player name-entry flow.

Color a Cube (shipped VR title)
Unity / C# | https://store.steampowered.com/app/3366710/ColorACube/

- ✓ Contributed to the commercial release of a VR painting game, supporting implementation, testing, debugging, and platform-specific improvements across Steam, Meta, and Pico.
- ✓ Implemented gameplay features including brush painting, hand-tracking paint input, and a replay system allowing players to rewatch their creative process.

StrikeMania (Steam)
Unity / C# | https://store.steampowered.com/app/5251980/Strikemania/

- ✓ Pooled missiles and VFX, trimmed allocations and skipped unchanged renderer state, measured by an editor stress benchmark.
- ✓ Shipped the display settings: Performance, Balanced and Quality presets over anti-aliasing, bloom and scanline toggles, resolution and window mode, non-16:9 support and GPU detection for laptop defaults.

Shiver Thy Timbers (GDWC 2024 Finalist)
Unity / C# |https://wakolat.itch.io/shiver-thy-timbers

- ✓ Built AI behaviour systems using a self-made behaviour tree framework, enabling modular pirate behaviour authoring and in-editor debugging by non-engineers.
- ✓ Integrated navigation and state-driven NPC decision-making supporting work/rest/wander and reactive threat behaviours across common and special pirate roles.

Key Skills

C++ | C# | Gameplay programming | Platform porting support | Object-oriented programming | Design patterns | Data structures | Algorithms | Debugging | Logging | Refactoring

Technical Proficiencies

Rider | Unity | Unreal Engine | YouTrack | Gitea | CMake | JSON | SQL | Notion | React | Sentry | Supabase

Education

BSc Game Development – Digital Arts and Entertainment (cum laude)
Howest, Kortrijk, Belgium | 2022 – 2026

Associate Degree, Programming
Artesis Plantijn, *Antwerp, Belgium, 2020-2022*